



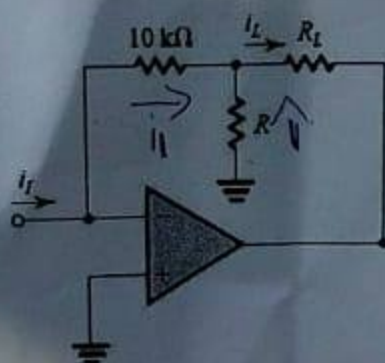
Course: Electronic Circuit Design
Program: BS(Electrical Engineering)
Duration: 60 Minutes
Paper Date: 01-Oct-2018
Section: ALL
Exam: Midterm-I

Course Code: EE224
Semester: Fall 2018
Total Marks: 30
Weight: 15%
Page(s): 1
Questions: 3

- Q1 A current amplifier for which $R_i = 1\text{ k}\Omega$, $R_o = 10\text{ k}\Omega$, and $A_{is} = 100\text{ A/A}$ is to be connected with a 10 mA source with a source resistance of $100\text{ k}\Omega$ and a load-resistance of $1\text{ k}\Omega$. What is the value of overall current gain i_L/i_s , expressed in ratio and in dB. [10] CLO 01

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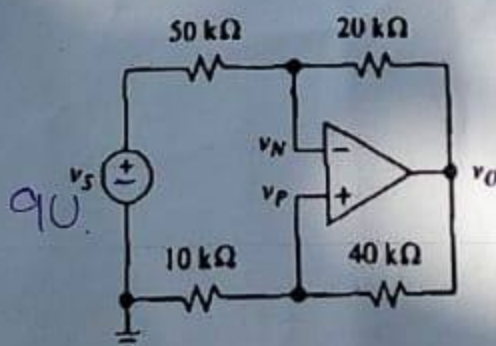
- Q2 Assuming the op-amp to be ideal, it is required to design the circuit shown in the following figure to implement a current amplifier with gain $i_L/i_f = 20\text{ A/A}$. Find the required value of R . [10] CLO 02



$$v_o = -\frac{R_i}{R_f} \left(1 + \frac{R_{c1}}{R_2} + \frac{R_{c4}}{R_3} \right)$$

$$i_f + i_3 = i_L$$

- Q3 Using the ideal operational amplifier in the following circuit for $v_s = 9\text{ V}$, find v_P , v_N , and v_O ? [10] CLO 02



$$v_P = 9V$$



Course: Electronic Circuit Design
 Program: BS(Electrical Engineering)
 Duration: 180 Minutes
 Paper Date: 18-Dec-2018
 Section: ALL
 Exam: Final

Course Code: EE224
 Semester: Fall 2018
 Total Marks: 100
 Weight: 50%
 Page(s): 3
 Questions: 5

Q

With the emitter used as a common terminal between input and output, Fig. 1 shows a transistor amplifier known as a **common-emitter** or **grounded-emitter** circuit. The circuit parameters are: $R_s = 5 \text{ k}\Omega$, $r_\pi = 2.5 \text{ k}\Omega$, $g_m = 40 \text{ mA/V}$, $r_o = 100 \text{ k}\Omega$, and $R_L = 5 \text{ k}\Omega$.

- Derive an expression for the voltage gain v_o/v_s .
- Evaluate the magnitude of gain.
- Evaluate the magnitude of gain if the effect of r_o is neglected.
- An alternative model for the transistor in which the trans-conductance amplifier is replaced by a current amplifier is shown in Fig. 2. Evaluate the short-circuit current gain β .

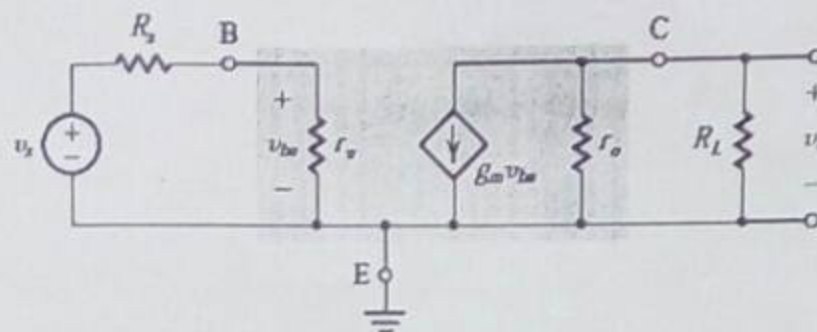


Fig. 1

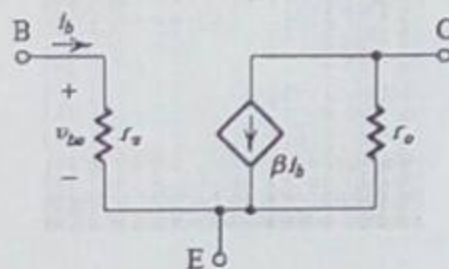


Fig. 2

- Q2 a) An internally compensated op-amp has an open-loop voltage gain of 80 dB. The corner frequency occurs at 6 Hz. [20]
- Find the unity gain frequency (0 dB frequency).
 - If the same op-amp is now connected in a closed-loop to form an inverting amplifier with a closed-loop gain of $G = -9 \text{ V/V}$. Find the corner frequency for this closed-loop amplifier.

- b) Design a circuit using only one op-amp that provides output signal, given by Eq. 1, for an input signal of V_i . Only $100.0 \text{ k}\Omega$ resistors are to be used in the circuit design.

$$V_o = -V_i - 0.01 \frac{dV_i}{dt} \quad (1)$$

- Clearly construct the circuit schematic.
- Find the values of all the components used in the circuit.

[20]

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1.1

- Q3 The 555 timer circuit shown in Fig. 3 is an astable multi-vibrator.
- Write down the expressions for the time-period and the duty-cycle of the output waveform if a diode is connected anti-parallel to R_B .
 - Hence, design the values of R_A and R_B so that the circuit can generate a perfect square-wave signal (50% duty-cycle) having a 1000 Hz frequency. Use a $0.1 \mu\text{F}$ capacitor in the design.
 - Construct the diagram of an alternative circuit that uses only one op-amp, instead of 555 timer, to generate a perfect square wave of 1000 Hz using a capacitor of $0.1 \mu\text{F}$. Find the unknown resistances if $\beta = 0.5$. The maximum value of resistance allowed in this circuit is $1.0 \text{ M}\Omega$.

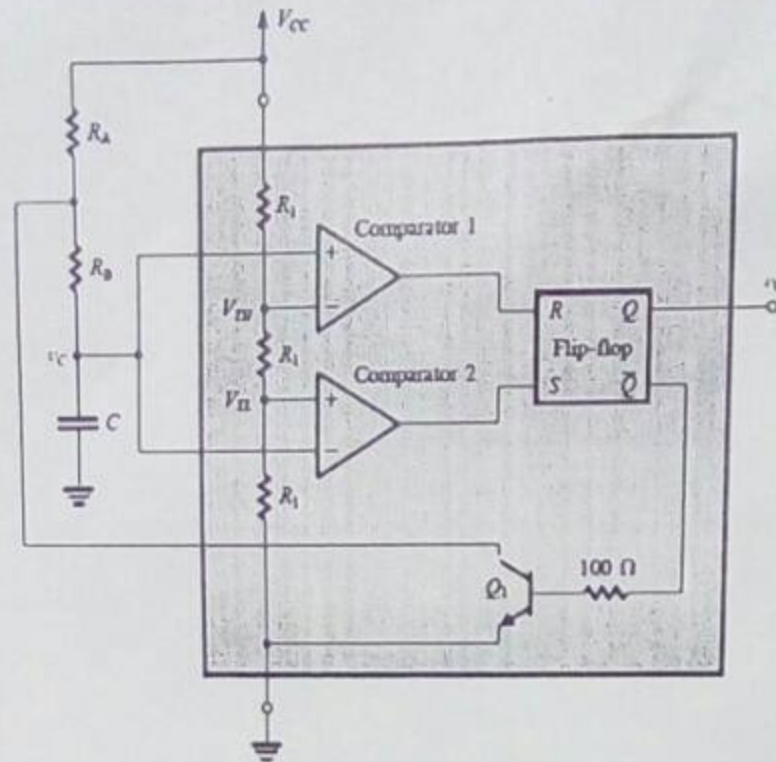


Fig. 3

- Q4 The circuit shown in Fig. 4 provides transfer characteristics with $\pm 7.5 \text{ V}$ levels at the output v_o . When $v_i = 0 \text{ V}$, a current of 0.5 mA flows in the feedback resistor (R_2) and a current of 1.0 mA flows through the zener-diodes. Assume that the output saturation levels of the op-amp are $\pm 12.0 \text{ V}$, the forward-drop voltage of the zener-diode is 0.7 V , and the value of R_1 is equal to R_2 .
- Specify the zener-voltages of the zener diodes.
 - Find the values of all resistors.

[20]

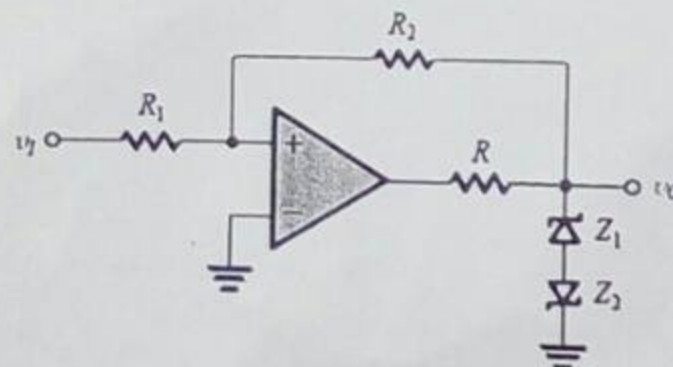
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Fig. 4

- Q5 Clearly draw the CMOS implementation of $Y = AB(C+D)$ using:
- NAND and NOR gates (draw the combinational logic circuit diagram as well)
 - Post-inversion technique
- Do not use the pre-inversion technique!*

[20]

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